Lecture L12

*Blind review — please score this lecture on row **L12** in the Excel form (1–10 per dimension).*

Normalization Forms and Database Schema Design (1NF-BCNF)

Learning Objectives:

- Explain the purpose of normalization in database schema design.
- Describe the five normal forms used in database schema design, including 1NF, 2NF, 3NF, 4NF, and BCNF.
- Explain the requirements of each normal form.
- Describe the benefits of each normal form.

Normalization Forms

Normalization is the process of organizing data in a database to minimize redundancy and dependency. It is essential to ensure the data is stored in an efficient manner and can be easily accessed, modified, and queried.

1NF (First Normal Form)

1NF requires that each column in a table contains atomic values, meaning that each value in the column is an individual piece of data. This means that each value in the column is a separate record or row. For example, in a table of customers, each customer's name would be a separate record.

$$\text{CustomerID} \rightarrow \text{Customer Name}$$

2NF (Second Normal Form)

2NF requires that a table is in 1NF and all non-key attributes are fully functionally dependent on the primary key. This means that each non-key attribute depends on the primary key to determine its value. For example, in the above customer table, the customer's address would depend on the customer's ID.

$$\text{CustomerID} \rightarrow \text{Customer Name} \rightarrow \text{Customer Address}$$

3NF (Third Normal Form)

3NF requires that a table is in 2NF and all non-key attributes are not transitive, meaning that they do not depend on each other to determine their values. In other words, the non-key attributes do not have a dependency on the non-key attributes. For example, in the customer table, the customer's ID and address are not dependent on each other.

$$\text{CustomerID} \rightarrow \text{Customer Name} \rightarrow \text{Customer Address}$$

4NF (Fourth Normal Form)

4NF requires that a table is in 3NF and all non-key attributes are not only not transitive, but also do not depend on each other. This means that the non-key attributes have no dependencies on each other. For example, in the customer table, the customer's ID and address are not dependent on each other.

$$\text{CustomerID} \rightarrow \text{Customer Name} \rightarrow \text{Customer Address}$$

BCNF (Boyce-Codd Normal Form)

BCNF is a higher normal form than 4NF, which is used in the design of hierarchical databases, where data is organized in a tree structure.

$$\text{CustomerID} \rightarrow \text{Customer Name} \rightarrow \text{Customer Address}$$

Benefits of Each Normal Form

- 1NF: Eliminates redundancy and improves data consistency.
- 2NF: Eliminates data anomalies and improves data integration.
- 3NF: Eliminates data inconsistencies and improves data redundancy.
- 4NF: Eliminates data anomalies and improves data redundancy.
- BCNF: Eliminates data inconsistencies and improves data redundancy.

By understanding these normal forms, database designers can create more efficient and reliable database schemas.